Merged dataset

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2026-09-27

```
# CARDIA
rmarkdown::render("/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 TB_DM/clean
files/2026.09.20 - V4/cardia.Rmd")

# NHANES
rmarkdown::render("/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 TB_DM/clean
files/2026.09.20 - V4/nhanes.Rmd")

# SOL
rmarkdown::render("/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 TB_DM/clean
files/2026.09.20 - V4/hchs_sol.Rmd")
```

Merging datasets

```
# categorical variables
df_list <- list(
  df__cardia_00    = df_cardia_00,
  df__cardia_10    = df_cardia_10,
  df__cardia_15    = df_cardia_15,
  df__cardia_20    = df_cardia_20,
  df__nhanes_1999  = df_nhanes_1999,
  df__nhanes_2011  = df_nhanes_2011,
  df__sol_v1       = df_sol_v1,
  df__sol_v2       = df_sol_v2
)

char_vars <- c(
  "id_study", "id", "drug_insu_db", "race", "sex", "edu_adult",
  "self_db", "drug_pill_db", "tb_treatment",
  "tb_skin_test_result", "tb_medication_use",
  "tb_history", "tb_skin_test", "self_lab_dm", "tb_dm"
)

library(dplyr)
library(purrr)

df_list <- map(df_list, function(df) {
  df %>%
    mutate(across(any_of(char_vars), as.character))
})

# numeric variables
numeric_vars <- c("age_told_db", "age_told_tb")

df_list <- map(df_list, function(df) {

  for (var in numeric_vars) {
    if (var %in% names(df)) {
```

```
      df[[var]] <- readr::parse_number(
        as.character(df[[var]]),
        na = c("Q", "S", "", "NA")
      )
    }
  }

  df
})

#checking
map(df_list, ~ sapply(.[char_vars], class))

list2env(df_list, envir = .GlobalEnv)

# cleaning
rm(df_list, df_cardia_00, df_cardia_10, df_cardia_15, df_cardia_20, df_nhanes_1999,
   df_nhanes_2011, df_sol_v1, df_sol_v2)

# Merging CARDIA, NHANES, SOL

library(dplyr)

merged_data <- bind_rows(
  df__cardia_00,
  df__cardia_10,
  df__cardia_15,
  df__cardia_20,
  df__nhanes_1999,
  df__nhanes_2011,
  df__sol_v1,
  df__sol_v2
)

# checking
table(merged_data$tb_dm) # n=137 + (updated: Y15 (3) + Y20(2))  => 142

# "Clean" dataset
merged_data <- merged_data %>%
  filter(tb_dm == "Yes")          # n=142
```

Preliminary Results

```
# Create table
merged_data_table <- merged_data %>%
  select(all_of(vars_to_summarize)) %>%
  tbl_summary(
    type = list(
  age_told_tb ~ "continuous",
  all_continuous() ~ "continuous"
),
    statistic = list(
      all_continuous() ~ "{median} ({p25}, {p75})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
```

```
    missing = "no"
  ) %>%
  modify_header(label ~ "**Variable**") %>%
  bold_labels() %>%
  as_gt() %>%
  gt::tab_header(
    title = "Summary of merged datasets"
  )

merged_data_table
```

Table 1: Summary of merged datasets

Variable	N = 142[1]
id_study	
CARDIA_Y10	4 / 142 (2.8%)
CARDIA_Y15	5 / 142 (3.5%)
CARDIA_Y20	19 / 142 (13%)
HCHS_SOL_V1	89 / 142 (63%)
HCHS_SOL_V2	17 / 142 (12%)
NHANES_1999_2000	3 / 142 (2.1%)
NHANES_2011_2012	5 / 142 (3.5%)
Age	51 (37, 60)
Sex	
Men	44 / 125 (35%)
Women	81 / 125 (65%)
Race	
Mexican/Other Hispanic	55 / 125 (44%)
Missing	36 / 125 (29%)
Non-Hispanic Black	31 / 125 (25%)
Non-Hispanic White	1 / 125 (0.8%)
Other/Multiracial	2 / 125 (1.6%)
Ethnicity	
Non-Hispanic	4 / 112 (3.6%)
Hispanic	108 / 112 (96%)
Missing	0 / 112 (0%)
Education Level	
< High School	31 / 142 (22%)
College+	5 / 142 (3.5%)
High School Grad	51 / 142 (36%)
Refused/DK	26 / 142 (18%)
Some College/AA	29 / 142 (20%)
Self-reported diabetes	100 / 125 (80%)
Age told DM	42 (34, 49)
Diabetes Mellitus on insulin	23 / 80 (29%)
Diabetes Mellitus on Oral med	
No	31 / 96 (32%)
Refused/DK	1 / 96 (1.0%)
Yes	64 / 96 (67%)
DM self-reported and by lab def	
Diabetes	139 / 141 (99%)
No Diabetes	2 / 141 (1.4%)
Family history of DM	1 / 1 (100%)
TB History ever	
Yes	142 / 142 (100%)
Age told TB	27.00 (27.00, 33.00)
TB treatment ever	82 / 111 (74%)

Variable	N = 142[1]
Fasting (minutes)	796 (650, 860)
Fasting Glucose (mg/dl)	127 (104, 186)
HbA1c (%)	7.20 (6.40, 8.60)
Total cholesterol (mg/dL)	193 (162, 230)
LDL (mg/dL)	108 (83, 142)
Weight (kg)	79 (70, 94)
Height (cm)	161 (155, 167)
BMI (kg/m2)	31 (27, 36)
Waist (cm)	101 (93, 113)
Hip (cm)	107 (100, 119)
Systolic blood pressure (mmHg)	124 (115, 133)
Diastolic blood pressure (mmHg)	74 (67, 81)
TB & DM	
Yes	142 / 142 (100%)

[1]n / N (%); Median (Q1, Q3)

UNIQUE PARTICIPANTS

```
# Create table
merged_data_table_unique <- merged_data_unique %>%
  select(all_of(vars_to_summarize)) %>%
  tbl_summary(
    type = list(
  age_told_tb ~ "continuous",
  all_continuous() ~ "continuous"
),
    statistic = list(
      all_continuous() ~ "{median} ({p25}, {p75})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    missing = "no"
  ) %>%
  modify_header(label ~ "**Variable**") %>%
  bold_labels() %>%
  as_gt() %>%
  gt::tab_header(
    title = "Summary of merged datasets"
  )

merged_data_table_unique
```

Table 2: Summary of merged datasets

Variable	N = 116[1]
id_study	
CARDIA_Y20	19 / 116 (16%)
HCHS_SOL_V1	89 / 116 (77%)
NHANES_1999_2000	3 / 116 (2.6%)
NHANES_2011_2012	5 / 116 (4.3%)
Age	53 (44, 61)
Sex	
Men	44 / 116 (38%)
Women	72 / 116 (62%)
Race	
Mexican/Other Hispanic	55 / 116 (47%)
Missing	36 / 116 (31%)
Non-Hispanic Black	22 / 116 (19%)
Non-Hispanic White	1 / 116 (0.9%)
Other/Multiracial	2 / 116 (1.7%)
Ethnicity	
Non-Hispanic	4 / 95 (4.2%)
Hispanic	91 / 95 (96%)
Missing	0 / 95 (0%)
Education Level	
< High School	31 / 116 (27%)
College+	5 / 116 (4.3%)
High School Grad	50 / 116 (43%)
Refused/DK	2 / 116 (1.7%)
Some College/AA	28 / 116 (24%)
Self-reported diabetes	91 / 116 (78%)
Age told DM	44 (35, 49)
Diabetes Mellitus on insulin	23 / 80 (29%)
Diabetes Mellitus on Oral med	
No	31 / 96 (32%)
Refused/DK	1 / 96 (1.0%)
Yes	64 / 96 (67%)
DM self-reported and by lab def	
Diabetes	115 / 116 (99%)
No Diabetes	1 / 116 (0.9%)
Family history of DM	1 / 1 (100%)
TB History ever	
Yes	116 / 116 (100%)
Age told TB	27.50 (27.00, 33.50)
TB treatment ever	71 / 97 (73%)
Fasting (minutes)	790 (45, 829)
Fasting Glucose (mg/dl)	125 (102, 181)
HbA1c (%)	7.15 (6.30, 8.35)

Variable	N = 116[1]
Total cholesterol (mg/dL)	193 (162, 231)
LDL (mg/dL)	108 (86, 138)
Weight (lbs)	80 (70, 93)
Height (cm)	161 (154, 167)
BMI (kg/m2)	31 (28, 36)
Waist (cm)	101 (96, 112)
Hip (cm)	106 (100, 117)
Systolic blood pressure (mmHg)	125 (115, 133)
Diastolic blood pressure (mmHg)	74 (67, 82)
TB & DM	
Yes	116 / 116 (100%)

[1]n / N (%); Median (Q1, Q3)

```
# Create table
merged_data_table_unique_study <- merged_data_unique %>%
  select(all_of(vars_to_summarize)) %>%
  tbl_summary(
    by = id_study,  # group by studies
    type = list(
  age_told_tb ~ "continuous",
  all_continuous() ~ "continuous"
),
    statistic = list(
      all_continuous() ~ "{median} ({p25}, {p75})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    missing = "no",
  ) %>%
  add_overall() %>%
  modify_header(label ~ "**Variable**") %>%
  bold_labels() %>%
  as_gt() %>%
  gt::tab_header(
    title = "Summary of merged datasets by Study"
  )

merged_data_table_unique_study
```

Table 3: Summary of merged datasets by Study

Variable	**Overall** N = 116[1]	**CARDIA_Y20** N = 19[1]	**HCHS_SOL_V1** N = 89[1]	**NHANES_1999_2000** N = 3[1]	**NHANES_2011_2012** N = 5[1]
Age	53 (44, 61)	26 (21, 28)	54 (49, 61)	67 (66, 74)	59 (54, 61)
Sex					
Men	44 / 116 (38%)	4 / 19 (21%)	35 / 89 (39%)	3 / 3 (100%)	2 / 5 (40%)
Women	72 / 116 (62%)	15 / 19 (79%)	54 / 89 (61%)	0 / 3 (0%)	3 / 5 (60%)
Race					
Mexican/Other Hispanic	55 / 116 (47%)	0 / 19 (0%)	53 / 89 (60%)	2 / 3 (67%)	0 / 5 (0%)
Missing	36 / 116 (31%)	0 / 19 (0%)	36 / 89 (40%)	0 / 3 (0%)	0 / 5 (0%)
Non-Hispanic Black	22 / 116 (19%)	18 / 19 (95%)	0 / 89 (0%)	1 / 3 (33%)	3 / 5 (60%)
Non-Hispanic White	1 / 116 (0.9%)	1 / 19 (5.3%)	0 / 89 (0%)	0 / 3 (0%)	0 / 5 (0%)
Other/Multiracial	2 / 116 (1.7%)	0 / 19 (0%)	0 / 89 (0%)	0 / 3 (0%)	2 / 5 (40%)
Ethnicity					
Non-Hispanic	4 / 95 (4.2%)	0 / 0 (NA%)	0 / 89 (0%)	1 / 3 (33%)	3 / 3 (100%)
Hispanic	91 / 95 (96%)	0 / 0 (NA%)	89 / 89 (100%)	2 / 3 (67%)	0 / 3 (0%)
Missing	0 / 95 (0%)	0 / 0 (NA%)	0 / 89 (0%)	0 / 3 (0%)	0 / 3 (0%)
Education Level					
< High School	31 / 116 (27%)	0 / 19 (0%)	28 / 89 (31%)	3 / 3 (100%)	0 / 5 (0%)
College+	5 / 116 (4.3%)	1 / 19 (5.3%)	0 / 89 (0%)	0 / 3 (0%)	4 / 5 (80%)
High School Grad	50 / 116 (43%)	8 / 19 (42%)	42 / 89 (47%)	0 / 3 (0%)	0 / 5 (0%)
Refused/DK	2 / 116 (1.7%)	1 / 19 (5.3%)	1 / 89 (1.1%)	0 / 3 (0%)	0 / 5 (0%)
Some College/AA	28 / 116 (24%)	9 / 19 (47%)	18 / 89 (20%)	0 / 3 (0%)	1 / 5 (20%)
Self-reported diabetes	91 / 116 (78%)	14 / 19 (74%)	72 / 89 (81%)	1 / 3 (33%)	4 / 5 (80%)
Age told DM	44 (35, 49)	38 (34, 43)	45 (38, 50)	43 (43, 43)	NA (NA, NA)
Diabetes Mellitus on insulin	23 / 80 (29%)	0 / 0 (NA%)	22 / 72 (31%)	1 / 3 (33%)	0 / 5 (0%)
Diabetes Mellitus on Oral med					
No	31 / 96 (32%)	2 / 13 (15%)	27 / 78 (35%)	1 / 1 (100%)	1 / 4 (25%)
Refused/DK	1 / 96 (1.0%)	0 / 13 (0%)	1 / 78 (1.3%)	0 / 1 (0%)	0 / 4 (0%)
Yes	64 / 96 (67%)	11 / 13 (85%)	50 / 78 (64%)	0 / 1 (0%)	3 / 4 (75%)
Family history of DM	1 / 1 (100%)	0 / 0 (NA%)	0 / 0 (NA%)	0 / 0 (NA%)	1 / 1 (100%)
TB History ever					
Yes	116 / 116 (100%)	19 / 19 (100%)	89 / 89 (100%)	3 / 3 (100%)	5 / 5 (100%)
Age told TB	27.50 (27.00, 33.50)	27.50 (27.00, 33.50)	NA (NA, NA)	NA (NA, NA)	NA (NA, NA)
TB treatment ever	71 / 97 (73%)	0 / 0 (NA%)	66 / 89 (74%)	2 / 3 (67%)	3 / 5 (60%)

Variable	**Overall** N = 116[1]	**CARDIA_Y20** N = 19[1]	**HCHS_SOL_V1** N = 89[1]	**NHANES_1999_2000** N = 3[1]	**NHANES_2011_2012** N = 5[1]
Fasting (minutes)	790 (45, 829)	810 (780, 847)	NA (NA, NA)	9 (7, 50)	14 (10, 39)
Fasting Glucose (mg/dl)	125 (102, 181)	113 (105, 173)	128 (99, 187)	109 (109, 109)	128 (107, 144)
HbA1c (%)	7.15 (6.30, 8.35)	NA (NA, NA)	7.15 (6.30, 8.35)	7.90 (6.50, 10.10)	6.40 (6.30, 7.40)
Total cholesterol (mg/dL)	193 (162, 231)	NA (NA, NA)	193 (162, 231)	NA (NA, NA)	NA (NA, NA)
LDL (mg/dL)	108 (86, 138)	87 (68, 108)	122 (89, 150)	187 (187, 187)	117 (61, 136)
Weight (lbs)	80 (70, 93)	86 (70, 108)	79 (71, 88)	96 (68, 100)	82 (66, 89)
Height (cm)	161 (154, 167)	164 (159, 170)	160 (153, 166)	166 (163, 166)	159 (158, 169)
BMI (kg/m2)	31 (28, 36)	31 (27, 42)	31 (28, 35)	36 (24, 36)	26 (26, 33)
Waist (cm)	101 (96, 112)	98 (85, 112)	102 (97, 110)	118 (96, 124)	100 (98, 113)
Hip (cm)	106 (100, 117)	NA (NA, NA)	106 (100, 117)	NA (NA, NA)	NA (NA, NA)
Systolic blood pressure (mmHg)	125 (115, 133)	119 (109, 133)	127 (115, 134)	125 (124, 131)	134 (114, 189)
Diastolic blood pressure (mmHg)	74 (67, 82)	77 (67, 83)	74 (67, 81)	62 (51, 76)	80 (70, 86)
TB & DM					
Yes	116 / 116 (100%)	19 / 19 (100%)	89 / 89 (100%)	3 / 3 (100%)	5 / 5 (100%)

[1]Median (Q1, Q3); n / N (%)